

Rahul Rawat

Data Scientist, sample preview

PERSONAL DETAILS

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GitHub	github.com/rahulrawat20022002
Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data scientist with hands on delivery in time series forecasting, recursive machine learning pipelines, and multi year front end development for a family owned engineering firm. Currently a Master of Science student in Data Science and Analytics at SRH University of Applied Sciences Heidelberg with **20 hours** per week Werkstudent availability immediately and full time from April 2027.

PROFESSIONAL EXPERIENCE

Oct 2025 to Mar 2026 Heidelberg	eRay GmbH Data Scientist - During a six month eRay GmbH and SRH Heidelberg collaboration to forecast water quality on a German lake, tasked with covering four indicators over rolling horizons, built an end to end recursive time series pipeline for chlorophyll a, turbidity, pH, and dissolved oxygen, delivered as a production ready module the client can rerun on every new sensor drop. - Faced with model choice ambiguity for the forecasting task and the need to convey uncertainty to non technical stakeholders, benchmarked six candidates head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, then landed on CatBoost multi quantile regression, which produced asymmetric 80 percent prediction intervals that gave the client decision support under uncertainty. - Concerned that naive time series splits would inflate accuracy, tasked with making the evaluation defensible, enforced strict anti leakage rules across the pipeline, which surfaced the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors. Technologies: <i>Python, CatBoost, Prophet, scikit learn, MICE</i>
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Aug 2023 to Aug
2024
India

SS Engineers and Contractors

Junior Associate Software Developer

- With SS Engineers and Contractors running internal Data Dashboards, Analytics platforms, and Employee Portals used across the company, tasked with building and maintaining the front end features that day to day teams depended on, contributed React UI components across all three internal products, which stayed in daily use by internal teams across the company throughout the year in role.
- With a client relying on a legacy AngularJS app that had to sit inside an existing module federation setup alongside newer React micro frontends, tasked with porting the client's screens across without breaking the running product, ported around 8 routes from AngularJS to React inside the module federation shell over **4 months** of incremental releases, shipped with no production incidents during rollout.
- With the same client migration surfacing shared UI patterns and a session boundary between the legacy AngularJS side and the new React side, tasked with keeping the migration code reusable rather than one off, wrote a set of reusable React UI components and an auth compatibility shim that bridged the legacy session shape to the new React app, both of which were later picked up by other developers on a team of 4.
- As the internal platforms and the client migration were both moving in parallel and regressions were slipping through manual QA, tasked with adding automated coverage, added Playwright end to end tests across the internal platform work and the client migration work, covering the main user flows so regressions on those flows were caught before release rather than after.

Technologies: *React, module federation, Playwright, AngularJS, HTML5, CSS3*

Feb 2023 to Aug
2023
India

SS Engineers and Contractors

Front End Developer Intern

- During a six month internship at SS Engineers and Contractors, tasked with learning the codebase and then taking on small UI work across the internal Data Dashboards and Employee Portals under senior review, paired closely with senior developers to walk the codebase and then shipped focused UI components including charts, filters, and profile pages, iterating each one on code review feedback until it landed.
- With intern code going straight into internal products that other teams used every day, tasked with catching problems earlier in the loop, fixed bugs across the parts of the codebase I worked on, wrote tests to cover the code I contributed, and helped the QA team investigate new issues as they came in, so problems on my areas were caught in review or in automated tests rather than in QA.

Technologies: *React, HTML5, CSS3, Git, code review workflow*

PERSONAL PROJECTS

2026
Personal

Hybrid RAG Orchestrator with Agentic Routing: Python, LangChain, Llama 3.1 8b via Groq, ChromaDB

- For a personal RAG project that needed to answer from local PDFs, live web, and small talk without a hardcoded path, tasked with building the router, delivered a modular Retrieval Augmented Generation system on top of Llama **3.1 8b** via Groq and LangChain orchestration with a custom decision making router that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, and returns clean end to end responses on all three routes.
- With document knowledge needing to persist across sessions rather than being reloaded each run, tasked with adding a durable semantic memory, built a ChromaDB store with local persistence and HuggingFace MiniLM L6 v2 embeddings over the PDF and vector data, which returned consistent retrieval on cross session queries.

Technologies: *Python, LangChain, Llama 3.1 8b via Groq, ChromaDB, HuggingFace MiniLM L6 v2, Streamlit*

EDUCATION

Apr 2025 to Present Heidelberg	M.Sc. Data Science and Analytics SRH University of Applied Sciences Heidelberg, GPA 1.9
2019 to 2023 Greater Noida, India	Bachelor of Technology in Computer Science GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023 Greater Noida, India	Bachelor Thesis, Diabetes Prediction Using Machine Learning GL Bajaj Institute of Technology and Management, IEEE style paper - For a Bachelor thesis on diabetes prediction with a small clinical dataset of 768 patients, tasked with building a defensible model comparison the examiners could audit, built a full end to end machine learning pipeline comparing six classifiers with 10 fold cross validation and per model confusion matrices, delivering a model comparison that stood up in the thesis defence. - Spotting biologically impossible zero values in the source data that the original authors had overlooked, tasked with restoring data integrity before any model fit, applied IQR based outlier removal and proper imputation, which lifted the dataset from silently broken to a clean training input for every downstream model. Technologies: <i>Python, scikit learn, Pandas, Seaborn</i>
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CERTIFICATIONS

- NVIDIA, Building LLM Applications With Prompt Engineering, issued 12 November 2025.
- AWS Academy Graduate, AWS Academy Cloud Foundations, issued 15 July 2025.
- SAS Certified Specialist, Visual Business Analytics Using SAS Viya, issued 7 May 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026, Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1 in progress toward B2
Hindi	Native